ECE 2305 – Introduction to C Programming
Programming Tutorial 17
Summation

Write a C++ program using a FOR Loop to evaluate the following summation.

$$sum = \sum_{n=1}^{100} \frac{1}{n^2} = \frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \cdots + \frac{1}{100^2}$$

Input

```
// ECE 2305 - Tutorial 17

#include <iostream>
using namespace std;

int main() {
    double sum = 0;
    for (int i = 1; i <= 100; i++) sum += 1.0 / (i * i);
    cout << "The answer to the summation is = " << sum << endl;
    cout << endl;

    system("pause");
    return 0;
}
```

Output

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